Importance Weighted Autoencoders on Dynamically Binarized MNIST: Reproduction and Extension

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Part I

Reproduction of Importance Weighted Autoencoders

Abstract

This part documents a compute-conscious reproduction of the one-stochastic-layer Importance Weighted Autoencoder (IWAE) experiment of Burda et al. [3]. The reconstruction preserves the paper's generative model, recognition model, objective with 50 importance samples, dynamic MNIST training binarization, initialization, optimizer, and staged learning-rate schedule, while using a modern PyTorch and PyTorch Lightning implementation. After 121 training passes on an Apple M1 GPU, the model obtains a 5,000-sample test negative log-likelihood estimate of 87.804 nats and 25 active latent units. The latter matches the paper's reported value, whereas the likelihood remains 3.024 nats above its 84.78-nat result. This gap is interpreted in light of the present run using only 3.7% of the paper's 3,280-pass optimization budget. The result establishes a documented, reproducible IWAE baseline without claiming full numerical replication.

I.1 Introduction

Variational autoencoders (VAEs) combine amortized inference with reparameterized stochastic optimization [6, 7]. Their standard evidence lower bound (ELBO), however, may provide a weak training signal when the approximate posterior is restrictive. IWAE replaces the single-sample ELBO with an importance-weighted bound and empirically learns representations with more active latent variables [3].

The objective of this reproduction is not to maximize MNIST performance at arbitrary computational cost. It is to establish a traceable result that (i) remains close enough to the original experiment for an academically meaningful comparison, (ii) runs on a 16 GB Apple M1 or a modest Colab accelerator, and (iii) clearly separates faithful scientific choices from implementation-level modernization.

I.2 Importance-weighted variational inference

For observation x, latent variable z, generative model $p_\theta(x, z)$, and recognition model $q_\phi(z \mid x)$, draw $z_{1:K} \overset{\text{iid}}{\sim} q_\phi(z \mid x)$ and define

$$w_k = \frac{p_\theta(x, z_k)}{q_\phi(z_k \mid x)}. \tag{1}$$

The IWAE objective is

$$\mathcal{L}_K(x) = \mathbb{E}_{z_{1:K}}\left[\log\left(\frac{1}{K}\sum_{k=1}^{K} w_k\right)\right]. \tag{2}$$

Under the conditions given by Burda et al. [3], $\mathcal{L}_1 \leq \mathcal{L}_K \leq \log p_\theta(x)$ and the expected bound is nondecreasing in K. Thus $K > 1$ gives a tighter variational objective, although a tighter bound does not by itself guarantee a better finite-budget optimizer trajectory. We minimize the corresponding negative bound in nats.

I.3 Reconstruction design

I.3.1 Data protocol

The paper describes the standard 60,000/10,000 MNIST train/test split; the released code reserves 400 of the 60,000 training examples for validation [2, 3]. We follow that implementation-level 59,600/400/10,000 split. Each training image is converted to a Bernoulli sample on every retrieval, following the random-binarization protocol in the authors' implementation. Validation and test examples instead receive one seeded Bernoulli draw that remains fixed. This documented deviation reduces evaluation noise and makes repeated evaluations use identical observations, but it prevents a perfectly like-for-like comparison with every stochastic evaluation detail of the original code.

I.3.2 Architecture and initialization

Table 1 gives the complete architecture. Both networks are multilayer perceptrons with two 200-unit tanh hidden layers. The recognition network outputs the mean and log standard deviation of a 50-dimensional diagonal Gaussian. The decoder outputs 784 Bernoulli logits. Weights use Glorot uniform initialization and biases are initially zero, matching the original source [2, 4]. The decoder's output bias is then set to the empirical training-pixel log odds, as in the source experiment. This initialization is explicit because PyTorch's default `Linear` initialization is not Glorot uniform; explicitness here is experimental fidelity rather than redundant framework code.

Table 1: One-stochastic-layer IWAE architecture. The stochastic layer is the 50-dimensional latent variable; each network also contains two deterministic hidden layers.

Component	Specification
Input	$x \in \{0, 1\}^{784}$
Encoder	$784 \to 200 \to 200 \to (50 + 50)$
Approximate posterior	diagonal Gaussian
Decoder	$50 \to 200 \to 200 \to 784$
Likelihood	factorized Bernoulli
Activation	tanh
Parameters	425,284
Training particles	$K = 50$

I.3.3 Optimization and reproducibility

Table 2: Optimization, software, and hardware configuration for the completed IWAE run.

Setting	Value
Optimizer	Adam [5]
Initial learning rate	10^{-3}
Adam parameters	$(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-4}$
Batch size	20
Numeric precision	32-bit floating point
Learning-rate schedule	Multiply by $\gamma = 10^{-1/7}$ at cumulative passes $\{1, 4, 13, 40, 121, 364, 1093, 3280\}$
Scheduler implementation	PyTorch `MultiStepLR`; the completed checkpoint used an algebraically equivalent direct schedule whose logged rates match the paper
Global random seed	236
Software	PyTorch 2.13.0; Lightning 2.6.5; torchvision 0.28.0; TorchMetrics 1.9.0
Host system	macOS 15.7.7; Apple M1; 16 GB unified memory
Accelerator	Metal Performance Shaders (MPS)
Persisted artifacts	Checkpoints, hyperparameters, and per-epoch metrics

I.4 Evaluation protocol

Test negative log likelihood (NLL) is estimated with $-\mathcal{L}_{5000}$, evaluated in chunks of 100 particles to bound device memory. Chunking accumulates the same log-sum-exp as an unchunked estimate and therefore changes memory use, not the estimator. Lower NLL is better.

Following Burda et al. [3], latent dimension j is counted as active when

$$\mathrm{Var}_x\Big[\mathbb{E}_{q_\phi(z|x)}[z_j]\Big] > 0.01. \tag{3}$$

This diagnostic measures variation in posterior means across the data; it is not proof that every active coordinate is semantically disentangled.

I.5 Results

Table 3: Published target and completed reproduction. NLL values are in nats. The training and validation quantities are $-\mathcal{L}_{50}$ and must not be compared numerically with test $-\mathcal{L}_{5000}$.

Quantity	Paper	This run
Training passes	3,280	121
Train $-\mathcal{L}_{50}$	–	90.268
Validation $-\mathcal{L}_{50}$	–	90.400
Test $-\mathcal{L}_{5000}$	84.78	87.804
Active units	25	25
Wall-clock time	–	74.2 min

The optimization curves in Figure 1 fall rapidly and remain close after the early passes. Both recorded minima occur at pass 121. This supports checkpoint selection at the end of the available budget, but it does not establish convergence: the original schedule allocates another 3,159 passes.

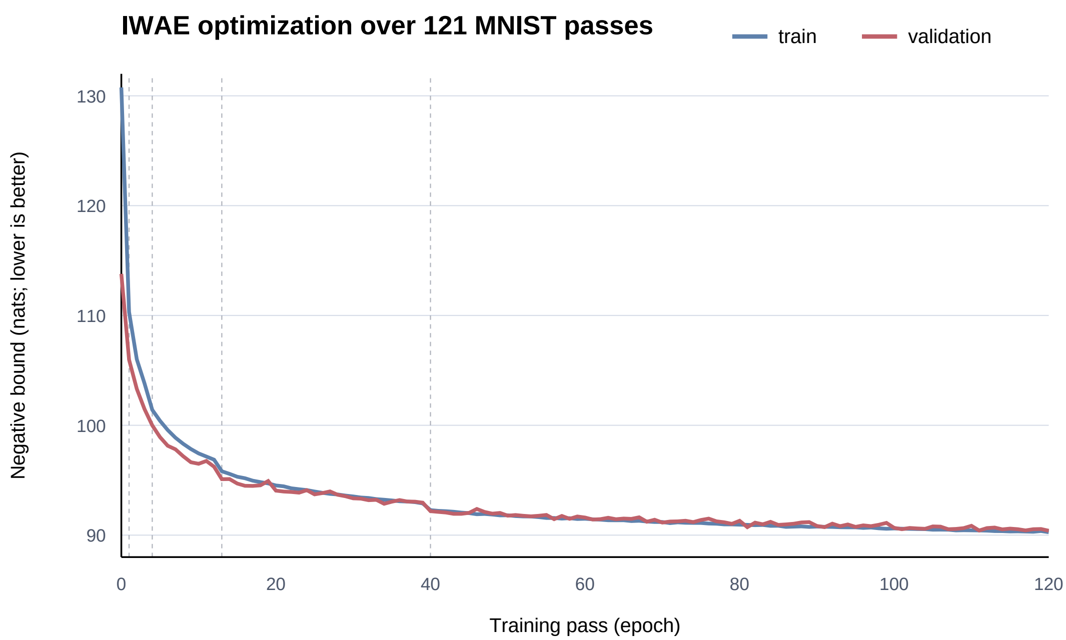

Figure 1: Epoch-level negative IWAE bounds for the completed 121-pass run. Dashed vertical lines denote learning-rate milestones within the observed window. Lower is better.

The 3.024-nat gap from the published test NLL is unsurprising under the truncated budget. The run consumes only $121/3280 = 3.69\%$ of the paper's training passes. Differences in the fixed evaluation draw, random-number generators, floating-point kernels, and framework implementation are secondary possible contributors. Because these factors are not independently ablated, the optimization budget is the strongest explanation, not a formally isolated cause.

The prior samples in Figure 2 are recognizably digit-like and cover several digit classes. Some strokes remain ambiguous or fragmented. This is useful qualitative evidence that the decoder learned the data domain, but sample inspection is not substituted for likelihood-based comparison.

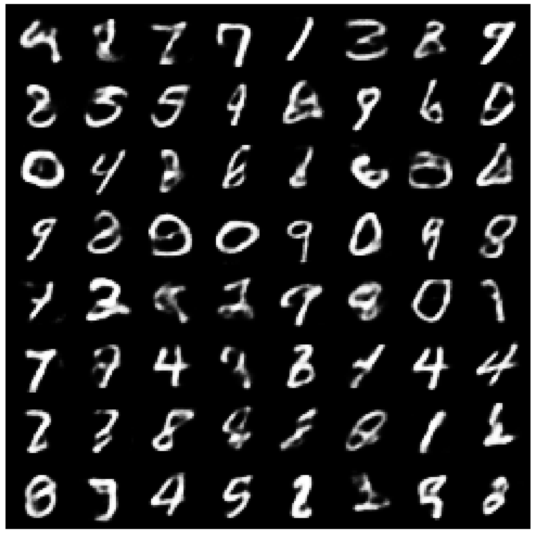

Figure 2: An 8×8 grid of prior predictive decoder probabilities from epoch index 120 (after 121 training passes). Brightness represents Bernoulli mean rather than a thresholded pixel draw.

I.6 Faithfulness and implementation modernization

The scientific core is preserved: random-binarized training data, architecture, latent size, $K = 50$ objective, initialization, optimizer hyperparameters, and milestone schedule. Device transfer, optimizer stepping, checkpoint serialization, metrics, and distributed-safe logging use maintained framework abstractions. Replacing those mechanics with handwritten equivalents would increase maintenance surface without increasing fidelity.

Three choices are intentionally modern. First, evaluation binarization is fixed per example for repeatability. Second, a chunked 5,000-particle calculation makes evaluation feasible on 16 GB unified memory. Third, the executable interface and saved configuration make every run self-describing. Only the first changes the stochastic evaluation protocol; the latter two are implementation-level changes.

I.7 Limitations

This is a single-seed, truncated-budget reproduction. Consequently it cannot estimate run-to-run uncertainty or claim numerical replication of the published NLL. MNIST likelihood also says little about behavior on natural images or text. The 74.2-minute duration is reconstructed from artifact timestamps rather than a trainer-emitted duration field, so it should be read as elapsed wall-clock time to timestamp precision. Finally, the active-unit threshold is inherited from the source study and is only one representation diagnostic.

I.8 Conclusion

The completed baseline demonstrates that the central one-stochastic-layer, 50-particle IWAE experiment is practical on an M1-class device. It reproduces the published active-unit count and produces coherent samples, while its likelihood gap is consistent with a deliberately much shorter training budget. The resulting checkpoint, metrics, and documented deviations form a credible and reproducible reconstruction of the selected paper condition.

Artifact availability

The run can be reproduced from the project root with:

```
uv run iwae train iwae -accelerator mps -max-epochs 121
```

The archived checkpoint is `outputs/iwae/checkpoints/epoch-0120.ckpt`; the test protocol uses:

```
uv run iwae evaluate outputs/iwae/checkpoints/epoch-0120.ckpt -accelerator mps
```

Part II

Doubly Reparameterized Gradient Study

Abstract

This part evaluates the doubly reparameterized gradient (DReG) estimator of Tucker et al. [8] as a controlled extension of the completed IWAE baseline. The data, architecture, initialization, generative objective, optimizer, particle count, training budget, and evaluation procedure are held fixed while the inference-network gradient estimator is changed. At the matched 121-pass budget, DReG obtains a test NLL of 87.480 nats and 26 active units, compared with 87.804 nats and 25 active units for IWAE. The 0.323-nat NLL reduction is a modest numerical improvement, but a single paired seed does not establish a reliable population-level advantage.

II.1 Motivation and hypothesis

As the number of importance samples grows, the conventional inference-network gradient estimator can suffer a deteriorating signal-to-noise ratio. DReG removes a high-variance score-function component by applying reparameterization a second time [8]. The working hypothesis is that, under an otherwise identical $K = 50$ experiment, this estimator will improve inference-network optimization without changing the model or generative objective.

II.2 Controlled experimental design

The comparison holds fixed the MNIST split and binarization draws, network architecture, initialization seed, $K = 50$, optimizer settings, learning-rate schedule, training budget, and evaluation implementation. The intended independent variable is the inference-network gradient estimator. Both conditions use seed 236 and 121 training passes, and each test result is computed from the checkpoint selected by validation NLL.

The primary comparison is test NLL estimated with $-\mathcal{L}_{5000}$. Active units will remain a secondary representation diagnostic. Kernel Inception Distance (KID) [1], computed in the fixed MNIST-classifier feature space, will provide a course-specific sample diagnostic. KID was not reported by the original IWAE paper and is comparable only when the feature extractor, real observations, subset configuration, sample count, and seed are identical.

II.3 Results

Table 4: Matched single-seed comparison after 121 training passes. Lower NLL is better. Active units are descriptive rather than an ordered quality metric.

Metric	IWAE	IWAE-DReG
Training passes	121	121
Selected checkpoint epoch	120	114
Train $-\mathcal{L}_{50}$ at selected epoch	90.268	89.652
Validation $-\mathcal{L}_{50}$ at selected epoch	90.400	89.661
Test $-\mathcal{L}_{5000}$	87.804	**87.480**
Active units	25	26
Active training time (min)	74.2	approximately 71.5
KID	not measured	not measured

Relative to IWAE, DReG reduces the estimated test NLL by 0.323 nats (approximately 0.37%) and increases the active-unit count by one. The DReG checkpoint was selected at epoch 114, where its validation bound was 0.739 nats lower than the IWAE checkpoint's validation bound. Both models were nevertheless allowed the same 121-pass training budget.

II.4 Discussion

The result is directionally favorable to DReG, but the effect is small relative to the remaining gap from the paper's 84.78-nat result. It is therefore more accurate to report a modest single-seed improvement than either a clear estimator win or no improvement. The active-unit change from 25 to 26 is also too small to support a strong representation claim.

The interrupted training sessions make the reported active training time an artifact-timestamp estimate, although it remains similar to the IWAE runtime. KID was not measured, so the likelihood result cannot be extended to a claim about sample quality. Matched repetitions with seeds 237 and 238 are needed to estimate variability and determine whether the NLL difference is consistent rather than seed-specific.

II.5 Conclusion

Under the completed seed-236 comparison, DReG improves test NLL from 87.804 to 87.480 nats while using one additional active latent unit and comparable training time. This is a measurable but modest improvement. Without replicated seeds or a sample-quality metric, the evidence does not justify a broad claim that DReG is reliably superior; it supports only a limited within-seed advantage under the present compute budget.

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